

HR Employee Attrition

Analysis & Interactive Dashboard

Python

SQL

Excel

Power BI

Prepared by: Almas Parwaiz

Data Analyst | HR Analytics Project



Executive Summary

End-to-End Workflow



Data Cleaning

Python

1



SQL Business Analysis

MySQL

2



KPI Reporting

Excel

3



Interactive Dashboard

Power BI

4

Key Outcome

Major factors driving attrition:



Overtime Workload



Low Salary



Job Satisfaction



Limited Career Growth

Business Problem & Objectives

Business Problem



Why are employees leaving?



Which employees are at risk?



Which departments have high turnover?



Impact: High recruitment cost · Loss of expertise · Low morale

Project Objectives



Analyze Attrition Patterns



Identify Key Turnover Factors



SQL Business Analysis



Build Interactive Dashboards



Provide HR Recommendations

Dataset Overview & Tools

Dataset Features

Age	Gender	Department
Job Role	Monthly Income	Overtime
Work-Life Balance	Years at Company	Job Satisfaction



Target Variable: Attrition (Yes:1 / No:0)

Tools & Technologies



Python

Data Cleaning & EDA



MySQL

Business Analysis



Excel

KPI Reporting



Power BI

Interactive Dashboard



GitHub

Project Hosting

Data Preparation & EDA

Data Preparation Steps

- 1 Removed duplicates
- 2 Handled missing values
- 3 Encoded categorical variables
- 4 Standardized column names
- 5 Created new features

Feature Engineering

Age Groups

Salary Bands

Risk Categories

Tenure Groups

And More

Key EDA Insights



Attrition by Department



Attrition by Salary Range



Overtime Impact on Attrition



Age Distribution vs Attrition



Gender-based Attrition

Visualizations: Bar Charts · Heatmaps · Distribution Plots

SQL Business Analysis

SQL Insights

Department-wise attrition

Salary of employees leaving

Overtime impact analysis

Age group risk analysis

Attrition trends over tenure

Advanced SQL Techniques



CTEs



Window Functions



Ranking Functions



Views



Stored Procedures

Key Business Insights



Overtime = Higher Attrition

Employees working overtime show significantly higher resignation rates



Low Salary = High Risk

Below-market compensation is a leading predictor of attrition



Sales Has Highest Turnover

Sales department consistently shows the highest attrition rate



Younger Employees Leave More

Employees under 30 have the highest attrition frequency



Poor Work-Life Balance

Low work-life balance scores correlate strongly with resignation

Power BI Dashboard & KPIs

Dashboard Features



KPI Cards



Filters & Slicers



Department Analysis



Salary Insights



Attrition Trends

Dashboard KPIs



1,470

Total Employees



237

Attrition Count



16.1%

Attrition Rate



\$-3.55E-16

Avg Salary



36.9 yrs

Average Age



5.13 yrs

Avg Tenure

HR Recommendations

Employee Retention



Reduce Overtime Workload

Limit mandatory overtime, enforce work-hour caps



Improve Salary Structure

Benchmark salaries against industry standards



Increase Engagement Programs

Regular recognition, team-building activities



Improve Work-Life Balance

Flexible schedules, remote options, wellness perks

HR Strategy



Monitor High-Risk Departments

Track Sales, R&D attrition with live dashboards



Conduct Satisfaction Surveys

Quarterly pulse surveys to detect dissatisfaction early



Improve Career Growth Plans

Mentoring, promotion tracks, upskilling programs



AI-Powered Predictions

Predictive models to flag at-risk employees proactively

Conclusion & Future Enhancements

Conclusion



Built complete HR analytics system using multiple tools



Identified key drivers of employee attrition



Created interactive dashboard for HR decision-making

Skills Demonstrated

Data Cleaning

SQL Analytics

Dashboard Design

BI & Visualization

Future Enhancements



ML Attrition Prediction

Predictive models for at-risk employees



Real-time HR Dashboard

Live data pipeline integration



AI-based Employee Insights

NLP on surveys & feedback



Cloud Database Integration

AWS / Azure cloud-hosted data



Automated Reporting

Scheduled email & BI reports